Supplementary Material for Controlled Reconstruction of Image-Based Time-Series Encodings: Operational Decoders for GAF, RP, and MTF Representations

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This document supplies extended results for the controlled encoder–decoder study in the main article. It uses the same notation, decoder contracts, data partitions, and distinction between normalised reconstruction and optional physical-unit calibration. No image generator is trained or evaluated here. The artificial signal used below is an analytical sanity check passed through the known encoders; it is not a sample from a generative model and provides no evidence about augmentation or synthetic-to-real transfer.

S1. Classifier configurations

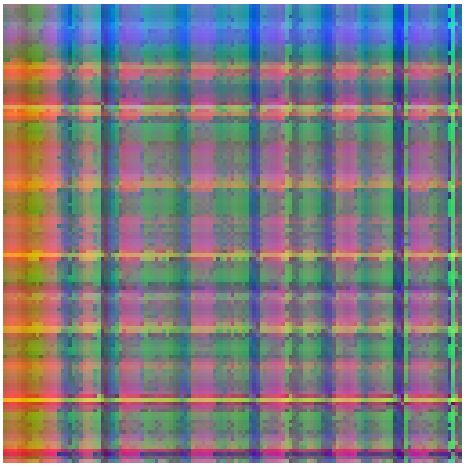
The classical baselines largely follow their library defaults, with the declared estimator count and seed for Random Forest. The recurrent and convolutional architectures are explicitly specified configurations rather than library defaults. No hyperparameter search used an outer-fold test prediction.

Table S1: Classifier configurations. Each model is trained once per fold on the original (un-reconstructed) windows and reused across all four encodings. Input is $(128, 3)$ for the neural models and the flattened 384-vector for the scikit-learn models; all neural models use the sparse categorical cross-entropy loss.

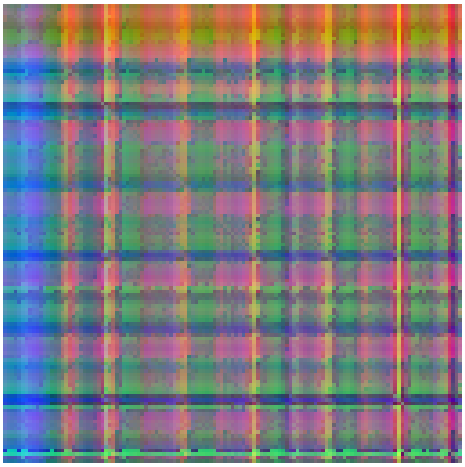
Classifier	Configuration	Seed
KNN	$k = 5$, uniform weights, Minkowski distance ($p = 2$)	—
SVM	RBF kernel, $C = 1.0$, $\gamma =$ scale	—
Random Forest	100 trees, Gini split, unlimited depth, `max_features`$= \sqrt{d}$	95
LSTM	LSTM(64) $\to$ Dropout(0.2) $\to$ LSTM(32) $\to$ Dropout(0.2) $\to$ Dense-softmax; Adam (10^{-3}), 20 epochs, batch 64	95
GRU	GRU(64) $\to$ Dropout(0.2) $\to$ GRU(32) $\to$ Dropout(0.2) $\to$ Dense-softmax; Adam (10^{-3}), 20 epochs, batch 64	95
CNN	Conv1D(196, kernel 16) + MaxPool(4) concatenated with per-channel statistical features (mean, variance, $\sum \lvert \cdot \rvert$, 10-bin histogram); Dense(1024) + Dropout(0.05) + Dense-softmax; $\ell_2 = 10^{-4}$; Adam (5×10^{-4}), 20 epochs, batch 64	95

S2. Encoding sensitivity to signal inversion

Before evaluating reconstructions, we first test whether each encoding visually distinguishes a time series from its sign-inverted counterpart. This check is important because an encoding that maps both signals to the same image cannot by itself resolve orientation during reconstruction. The same WISDM window (subject 1600, first walking window, $N = 128$) is used for every encoder so the only difference across panels is the encoder itself; the inverted variant negates all three accelerometer channels.

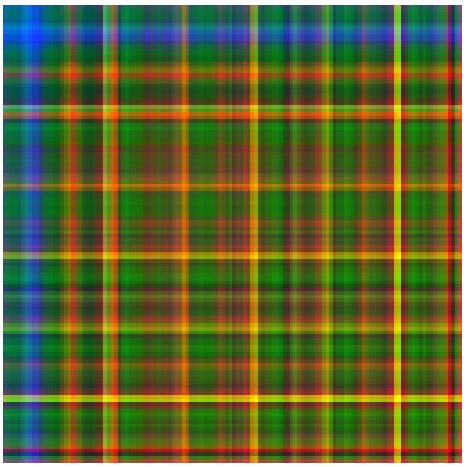


(a) Original

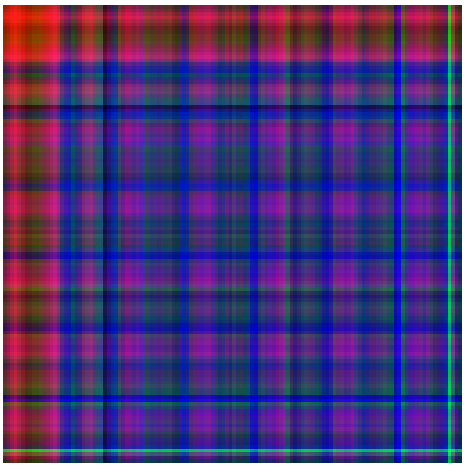


(b) Inverted

Figure S1: Recurrence Plot (RP) images: original vs. inverted.

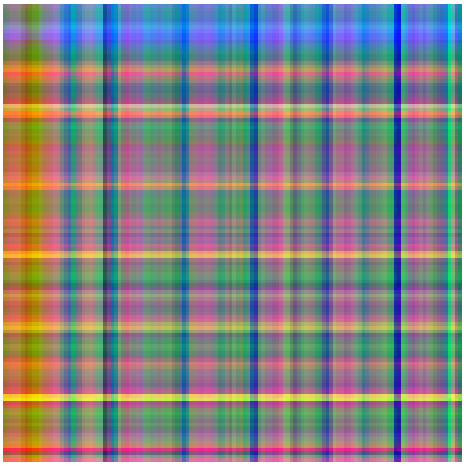


(a) Original

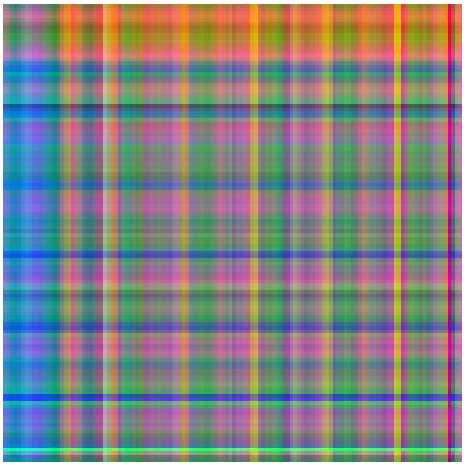


(b) Inverted

Figure S2: Gramian Angular Summation Field (GASF) images: original vs. inverted.

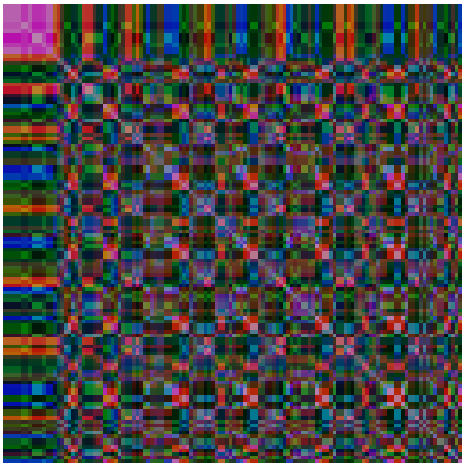


(a) Original

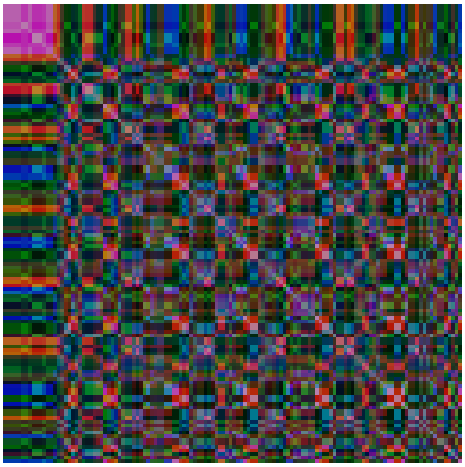


(b) Inverted

Figure S3: Gramian Angular Difference Field (GADF) images: original vs. inverted.



(a) Original



(b) Inverted

Figure S4: Markov Transition Field (MTF) images: original vs. inverted.

Figures S1 and S2 show that RP and GASF distinguish this window from its sign-inverted counterpart. The same holds for GADF (Figure S3): under the symmetric normalisation used here, negating the input flips the sign of $\sin(\phi_i - \phi_j)$ and therefore reverses the image contrast. Conversely, the MTF images in Figure S4 are identical. Under the ideal quantile relabelling $q \mapsto Q{+}1{-}q$, a global sign flip reverses both the bin order and the corresponding rows and columns of W, and these two permutations cancel in $M_{a,b} = W_{q_a,q_b}$. This operator-level sanity check illustrates the global-reversal ambiguity stated in Section 4.5; it does not by itself evaluate reconstruction fidelity, and tied quantiles can require additional implementation conventions.

S3. Additional qualitative diagnostics

S3.1. Analytical control signal

We construct an analytical three-channel signal $\mathbf{w} \in \mathbb{R}^{N\times 3}$ of length $N = 128$ by sampling an underlying parameter t uniformly over $[0, \pi]$. Concretely, define

$$t_i = \frac{i}{N-1}\,\pi, \qquad i = 0, 1, \ldots, N-1,$$

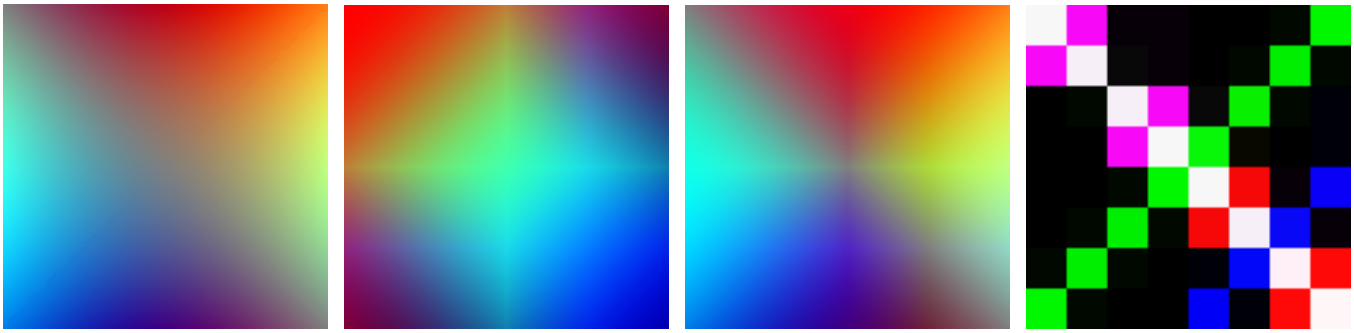
and let each vector $\mathbf{w}_i = [w_{i,0},\, w_{i,1},\, w_{i,2}]^\top$ be given by

$$w_{i,0} = \cos(t_i), \qquad w_{i,1} = \sin(t_i), \qquad w_{i,2} = \sin(t_i) \;-\; \cos(t_i).$$

Thus the full data matrix is

$$\mathbf{w} = \begin{bmatrix} \cos(t_0) & \sin(t_0) & \sin(t_0) - \cos(t_0) \\ \cos(t_1) & \sin(t_1) & \sin(t_1) - \cos(t_1) \\ \vdots & \vdots & \vdots \\ \cos(t_{N-1}) & \sin(t_{N-1}) & \sin(t_{N-1}) - \cos(t_{N-1}) \end{bmatrix}.$$

This signal, composed of a pure cosine channel, a pure sine channel, and their difference, serves as a controlled sanity check for the RP, GASF, GADF, and MTF reconstruction paths. It is not intended to represent the WISDM distribution or the output of a generative model; it provides a visually interpretable case where phase, shape, and orientation errors are easy to detect. Figure S5 shows the image encodings of the three channels (RP is 127×127 because of the $N - 1$ pair embedding; the others are 128×128).



(a) RP (b) GASF (c) GADF (d) MTF

Figure S5: Image encodings of the same analytical three-channel signal (cosine, sine, sine−cosine) under each technique.

After generating the encoded images, we decode them and compare the reconstructed series graphically.

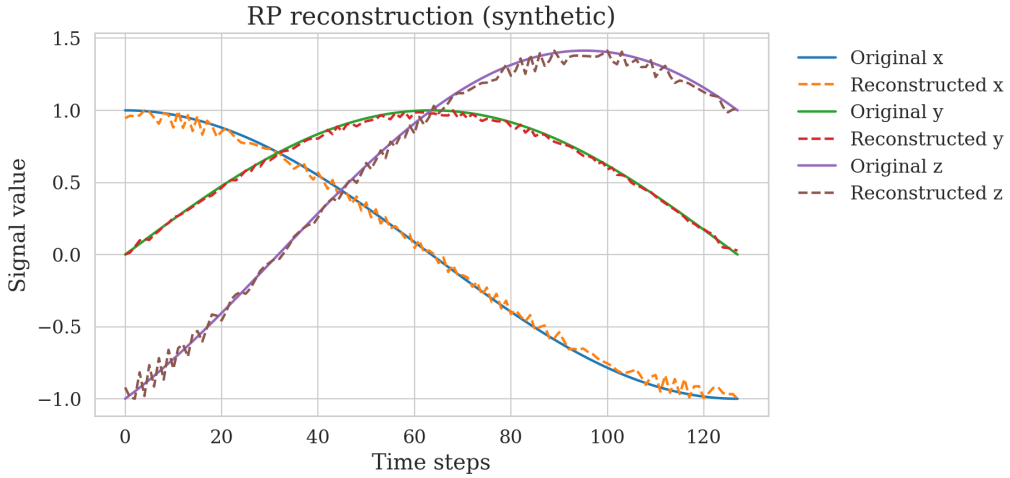


Figure S6: RP reconstruction of the analytical three-channel signal.

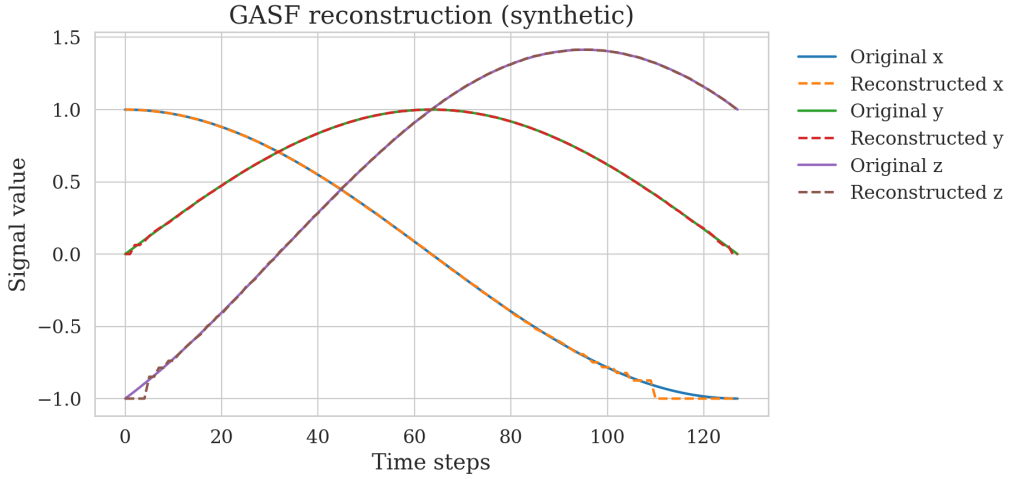


Figure S7: GASF reconstruction of the analytical three-channel signal.

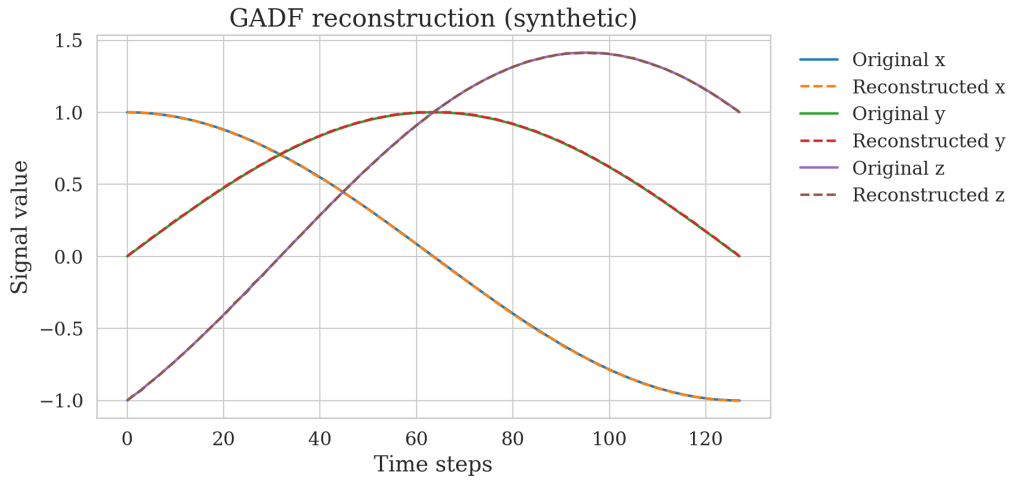


Figure S8: GADF reconstruction of the analytical three-channel signal.

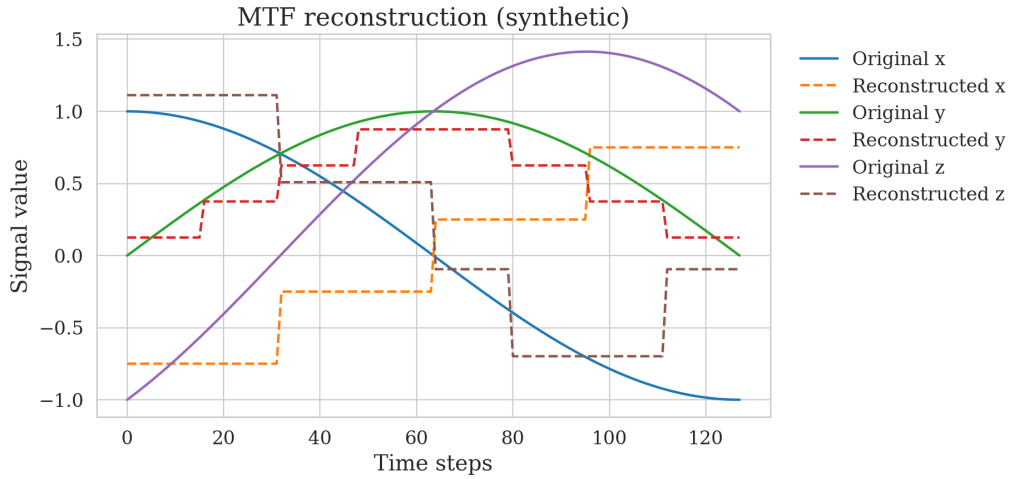


Figure S9: MTF reconstruction of the analytical three-channel signal.

After encoding the analytical signal (Figure S5), we reconstruct each representation and overlay it with the source. RP, GASF, and GADF visually follow the three normalised waveforms closely in Figures S6–S8. The plotted physical amplitudes use the same source-derived per-axis calibration as the main evaluation and therefore should not be interpreted as image-only recovery of physical scale. This single example illustrates the expected decoder behaviour but does not establish aggregate fidelity. MTF (Figure S9) instead exhibits quantised plateaus and the global-orientation ambiguity described above.

S3.2. Per-axis MTF view of the main WISDM example

The encoded images and four global reconstructions of WISDM walking window `1600×0` already appear in Figures 2 and 3 of the main article and are not duplicated here. Figures S10–S12 provide the additional per-axis view needed to see the quantile-level plateaus and ordering errors that are hidden by the global overlay. This example is qualitative only; aggregate claims rely on the full-dataset tables.

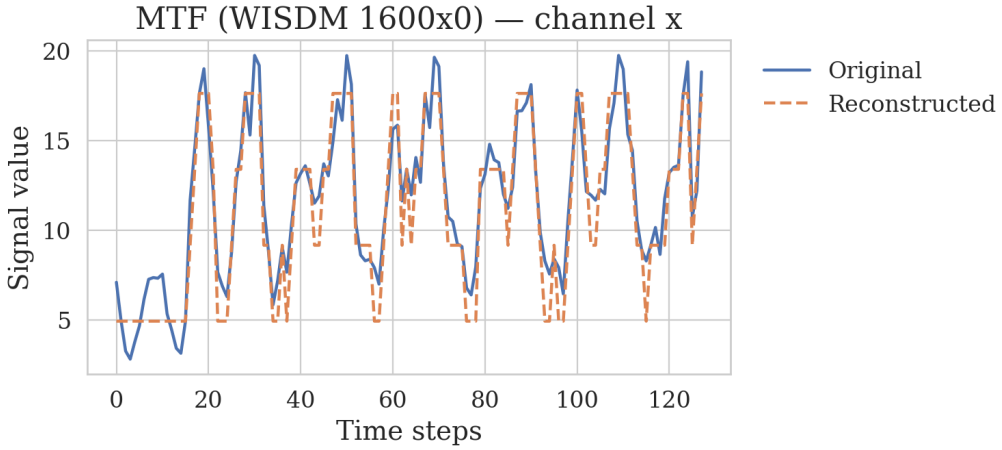


Figure S10: Original vs. MTF reconstruction for sample `1600×0`, axis x.

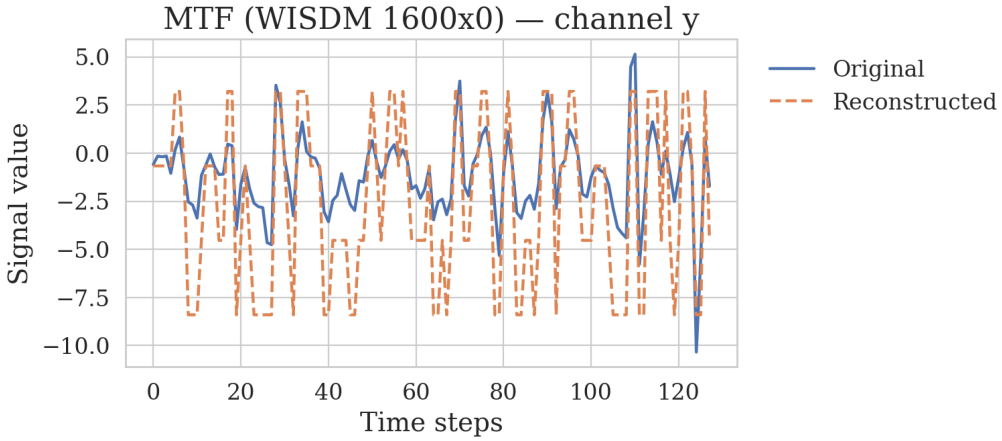


Figure S11: Original vs. MTF reconstruction for sample `1600×0`, axis y.

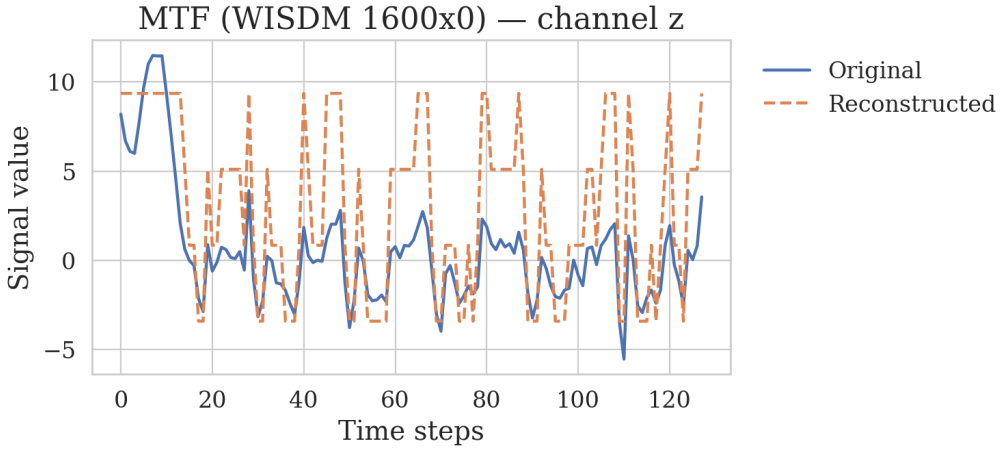


Figure S12: Original vs. MTF reconstruction for sample `1600×0`, axis z.

S4. Additional spectral-fidelity metrics

The main article separates spectral shape from total energy. Table S2 reports the complementary quantities omitted from the compact body table. Energy error is the relative difference in total Welch power; band-shape error is the mean absolute difference in normalised spectral mass over 0–1, 1–3, 3–5, and 5–10 Hz. Neither quantity is folded into the PSD-shape distance.

Table S2: Complementary spectral metrics over 8,172 windows (mean±std across windows, averaged over axes).

Method	Relative total-energy error	Band-shape error
RP	0.0030 ± 0.0027	0.00083 ± 0.00043
GASF	0.0054 ± 0.0055	0.00085 ± 0.00085
GADF	0.0011 ± 0.0009	0.00029 ± 0.00021
MTF	2.199 ± 8.844	0.05624 ± 0.03692

S5. Robustness details and error attribution

The local-stability sample is selected deterministically from window-level fold 0 (seed 0): 30 windows from each of the five activities, using round-robin sampling over the available subjects within each activity. The resulting 150 windows cover all 51 subjects. Fold 0 is only a reproducible sampling frame here: the decoders are untrained and no configuration is selected on this sample. For every nonzero noise level, two independent arrays are sampled over all pixels and channels, clipped to $[0, 255]$, rounded to the 8-bit grid, and decoded; replicate errors are averaged within each window. These two draws reduce replicate-specific variation but are not sufficient to characterise the full noise distribution at an individual window. Accordingly, the reported standard deviations and intervals summarise variability *across* windows, and the number of draws is exposed as `TFG_ROBUST_REPS` so the curve can be recomputed with more realizations. The subject-cluster intervals in Table S3 resample the 51 represented subjects as whole clusters. The MTF encoder emitted tied-quantile warnings for some axis-series; allowing fewer distinct bins in those cases is part of the evaluated quantisation behaviour.

Table S3: Subject-cluster bootstrap 95% intervals for mean NMAE along the perturbation curves (10^4 resamples). These accompany the between-window standard deviations in the main article.

Method	$\sigma = 0$	$\sigma = 2$	$\sigma = 4$	$\sigma = 8$	$\sigma = 16$
RP	$[0.0009, 0.0011]$	$[0.0039, 0.0049]$	$[0.0075, 0.0091]$	$[0.0153, 0.0183]$	$[0.0316, 0.0364]$
GASF	$[0.0014, 0.0015]$	$[0.0124, 0.0137]$	$[0.0207, 0.0222]$	$[0.0349, 0.0368]$	$[0.0621, 0.0647]$
GADF	$[0.0011, 0.0013]$	$[0.0071, 0.0077]$	$[0.0142, 0.0153]$	$[0.0304, 0.0327]$	$[0.0644, 0.0696]$
MTF	$[0.2686, 0.2950]$	$[0.2733, 0.2963]$	$[0.2752, 0.2991]$	$[0.2726, 0.2956]$	$[0.2694, 0.2903]$

The cycle residual is the relative Frobenius error defined in the main article and compares native float operator matrices, not PNG intensities. Table S4 also reports the maximum wiring-gate discrepancy over the sample. All four clean reconstructions agree numerically with their cached pipeline counterparts on every window; "passed" means the per-window maximum is below the stated 0.05 threshold, not that the threshold itself is an exactness claim.

Table S4: Self-consistency gate and clean image-domain cycle residual (mean±standard deviation and subject-cluster bootstrap 95% interval).

Method	Maximum gate NMAE	Cycle residual	Cycle 95% interval
RP	$< 10^{-8}$	0.0533 ± 0.0286	$[0.0497, 0.0569]$
GASF	$< 10^{-8}$	0.0046 ± 0.0012	$[0.0045, 0.0048]$
GADF	$< 10^{-8}$	0.0044 ± 0.0016	$[0.0042, 0.0046]$
MTF	$< 10^{-8}$	0.1242 ± 0.1955	$[0.0946, 0.1556]$

For the signed RP the cycle residual must not be read as a reconstruction-error measure. The Lu–Tong sign $\sigma(i, j)$ is a threshold on $\hat{u}_{ij} \cdot v$ at $\cos(\pi/4)$, so a pair whose displacement direction lies on that boundary flips sign under a perturbation of the order of double-precision rounding, and its entry then changes by $2|R_{ij}|$. Decoding the three axes of three windows directly from the native float operator reproduces the source signal to $\max_i |x_i - \hat{x}_i| \leq 7 \times 10^{-15}\,\mathrm{m/s^2}$, and yet the per-channel cycle residual ranges from exactly 0, for channels in which no pair flips, to 0.036. The RP row of Table S4 is therefore dominated by boundary flips, and it is not comparable with the GASF, GADF and MTF rows, whose operators are continuous in the signal.

The per-source error decomposition (native float operator, clean 8-bit PNG, and perturbed PNG at $\sigma = 8$) over these 150 windows is reported in the main text, Table 5. It shows that RP, GASF, and GADF are all exact in floating point, so their clean error is entirely 8-bit quantisation, whereas MTF is already lossy before storage. RP's earlier residual float term came from the angular orientation grid and disappears with the closed-form orientation (Section S11).

S6. Complete reconstruction results by fold

Table S5: Per-fold reconstruction error for both 3-fold protocols (physical units, m/s^2). Each cell is the per-window global metric (averaged over the three axes) averaged over that fold's test windows. Because the decoders are deterministic and untrained, a window reconstructs identically in any fold; the two protocols only repartition the same 8172 windows, so the Pooled column is identical for both and equals the full-dataset value of Table 2. The window-level folds are near-identical stratified partitions (negligible fold-to-fold spread), whereas the subject-out folds vary more because different subjects are held out.

		Window-level 3-fold			**Subject-out 3-fold**			
Encoder	**Metric**	F0	F1	F2	F0	F1	F2	**Pooled**
RP	MAE	0.0141	0.0143	0.0143	0.0133	0.0151	0.0143	0.0142
	MSE	0.0007	0.0009	0.0009	0.0007	0.0010	0.0008	0.0008
	Pearson	0.99998	0.99998	0.99998	0.99999	0.99998	0.99998	0.99998
GASF	MAE	0.0202	0.0205	0.0207	0.0189	0.0211	0.0214	0.0205
	MSE	0.0042	0.0043	0.0044	0.0037	0.0041	0.0051	0.0043
	Pearson	0.99988	0.99988	0.99988	0.99988	0.99988	0.99988	0.99988
GADF	MAE	0.0156	0.0157	0.0159	0.0147	0.0162	0.0163	0.0157
	MSE	0.0008	0.0008	0.0008	0.0007	0.0008	0.0009	0.0008
	Pearson	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999	0.99999
MTF	MAE	3.98	3.94	3.91	3.71	4.04	4.07	3.94
	MSE	56.42	55.25	54.62	48.13	52.54	65.43	55.43
	Pearson	0.00289	0.02089	0.01689	0.00977	0.00116	0.02982	0.01356

S7. Complete grouped subject-out classifier results

Table S6: Grouped subject-out three-fold accuracies for all six classifiers (mean±standard deviation across folds). Folds 0, 1 and 2 hold out disjoint groups of 17, 18 and 16 test subjects.

Classifier	Original	RP	GASF	GADF	MTF
KNN	0.504 ± 0.035	0.504 ± 0.035	0.504 ± 0.034	0.505 ± 0.034	0.465 ± 0.039
SVM	0.738 ± 0.020	0.738 ± 0.020	0.738 ± 0.020	0.738 ± 0.020	0.665 ± 0.026
RF	0.786 ± 0.039	0.786 ± 0.039	0.787 ± 0.039	0.787 ± 0.039	0.689 ± 0.026
LSTM	0.744 ± 0.029	0.744 ± 0.028	0.744 ± 0.029	0.744 ± 0.028	0.672 ± 0.035
GRU	0.811 ± 0.026	0.811 ± 0.026	0.811 ± 0.026	0.811 ± 0.026	0.702 ± 0.025
CNN	0.796 ± 0.036	0.796 ± 0.035	0.796 ± 0.036	0.796 ± 0.036	0.686 ± 0.035

S8. Window-level vs subject-out accuracy

Table S7 places the window-level (W, within-distribution) and grouped subject-out (S, primary) accuracies side by side for the three representative classifiers reported in the main text. Absolute accuracy is higher under the window-level split, as expected when windows of a test subject may also appear in training, but the encoder ranking is identical under both protocols: RP, GASF and GADF track the original signal, while MTF degrades every classifier. The main text reports the subject-out column only.

Table S7: Window-level (W) vs grouped subject-out (S) three-fold accuracy (mean across folds) for three representative classifiers. The encoder ranking is identical under both protocols.

	Original		RP		GASF		GADF		MTF	
Classifier	W	S	W	S	W	S	W	S	W	S
CNN	0.907	0.796	0.906	0.796	0.907	0.796	0.907	0.796	0.773	0.686
GRU	0.923	0.811	0.923	0.811	0.923	0.811	0.923	0.811	0.765	0.702
RF	0.891	0.786	0.891	0.786	0.891	0.787	0.892	0.787	0.757	0.689

S9. Subject-level classification equivalence

Simultaneous max-deviation bootstrap algorithm. The central equivalence result is a joint subject-cluster bootstrap. Because it is the headline statistic, we state the exact procedure. Let a *family* be a set of C comparisons (here $C = 18$ for the primary RP/GASF/GADF family, $C = 6$ for the separate MTF control family), each a fixed (classifier, encoder) pair, evaluated over the same $K = 51$ subjects in a fixed order.

1. **Per-subject differences.** For subject s and comparison c, compute the per-subject accuracy difference $\Delta_{c,s} = \mathrm{acc}_s^{\mathrm{orig}} - \mathrm{acc}_s^{\mathrm{recon}(c)}$, where each subject accuracy is that subject's fraction of correctly classified held-out windows (under subject-out validation every subject is a test subject in exactly one outer fold). This gives a $K \times C$ matrix with one row per subject.
2. **Point estimates.** The estimate for comparison c is the subject mean $\hat{\mu}_c = \frac{1}{K}\sum_s \Delta_{c,s}$ (the "mean Δacc" column). Subjects, not windows, are the sampling unit; no per-subject weighting or standardisation is applied.
3. **Joint resampling.** For $b = 1, \ldots, B$ ($B = 10^4$) draw one index vector of K subjects with replacement and apply the *same* vector to every column, so whole subjects are resampled jointly and the dependence between classifiers and encodings is preserved. Compute the resampled means $\hat{\mu}_c^{(b)}$ for all c.
4. **Family-wise critical value.** Centre each resampled mean on its point estimate, $d_c^{(b)} = \hat{\mu}_c^{(b)} - \hat{\mu}_c$ (raw accuracy units; the deviations are *not* studentised or divided by a bootstrap standard error), take the family maximum $D^{(b)} = \max_c |d_c^{(b)}|$, and set the single critical value κ to the 95th percentile of $\{D^{(b)}\}_{b=1}^{B}$. One κ is shared by all comparisons in the family, which is what makes the coverage simultaneous rather than pointwise.
5. **Intervals and decision.** The simultaneous interval for comparison c is $[\hat{\mu}_c - \kappa,\ \hat{\mu}_c + \kappa]$. Equivalence at margin m is declared iff that whole interval lies inside $[-m, +m]$; the main text uses $m = 0.01$ with $m = 0.02$ as a less stringent sensitivity check.

Each family uses an independent random seed and its own κ; no resamples or critical values are shared between the primary and MTF families, and none are pooled across the subject-out and window-level protocols. The compact main-text table reports, for each method, the single classifier-specific interval whose endpoint lies closest to the equivalence margin. The wider "interval spans" used later in Table S9 are unions of the classifier-specific intervals and are descriptive only. This is the `_cluster_bootstrap_simultaneous` routine in `scripts/_stat_tests.py`; the confirmatory paired TOST uses the same per-subject $\Delta_{c,s}$ but a Student-t interval at the bilateral 90% level, so the simultaneous 95% bootstrap is the stricter headline rule.

Table S8 exposes all classifier-specific simultaneous intervals behind the compact worst-case summary in the main article. The primary family contains the 18 RP/GASF/GADF comparisons; MTF's six negative-control comparisons form a separate family. Within each family, all classifiers share the same 10^4 subject resamples and the same bootstrap maximum-deviation critical value. Thus these are simultaneous intervals, not six unadjusted pointwise intervals and not an "envelope CI."

Table S8: Mean per-subject Δaccuracy and simultaneous 95% interval, Original − reconstruction, under grouped subject-out validation (51 subjects). Bold cells lie wholly inside $[-0.01, +0.01]$.

Classifier	RP	GASF	GADF	MTF
KNN	**−0.0000 [−0.0022, +0.0021]**	**−0.0001 [−0.0022, +0.0021]**	**−0.0005 [−0.0027, +0.0016]**	+0.0474 [+0.0212, +0.0736]
SVM	**+0.0004 [−0.0018, +0.0025]**	**+0.0000 [−0.0022, +0.0022]**	**+0.0008 [−0.0013, +0.0030]**	+0.0651 [+0.0389, +0.0913]
RF	**+0.0006 [−0.0016, +0.0027]**	**−0.0006 [−0.0027, +0.0016]**	**−0.0002 [−0.0024, +0.0019]**	+0.0910 [+0.0649, +0.1172]
LSTM	**+0.0005 [−0.0017, +0.0026]**	**−0.0000 [−0.0022, +0.0021]**	**+0.0001 [−0.0021, +0.0023]**	+0.0697 [+0.0435, +0.0959]
GRU	**−0.0004 [−0.0025, +0.0018]**	**−0.0002 [−0.0023, +0.0020]**	**−0.0003 [−0.0024, +0.0019]**	+0.1124 [+0.0862, +0.1386]
CNN	**+0.0003 [−0.0019, +0.0024]**	**+0.0003 [−0.0019, +0.0025]**	**+0.0002 [−0.0020, +0.0023]**	+0.1144 [+0.0882, +0.1405]

The Holm-adjusted paired TOST analysis gives the same method-level decision at the ±0.01 margin: equivalence is established for 6/6 classifiers with RP, GASF, and GADF, and for 0/6 with MTF. This confirmation is secondary to the simultaneous subject-cluster intervals above.

For completeness, Table S9 applies the same simultaneous procedure to a separate 18-comparison family under the secondary window-level protocol (and a separate six-comparison MTF family). No interval is pooled across protocols. Both protocols yield the same equivalence decisions.

Table S9: Descriptive spans of classifier-specific simultaneous 95% intervals by protocol. Each protocol is analysed as a separate family; the displayed span is not itself a confidence interval.

	Window-level (secondary)			Subject-out (primary)		
Method	Mean Δ range	Interval span	Eq. ±0.01	Mean Δ range	Interval span	Eq. ±0.01
RP	$[-0.0000, +0.0004]$	$[-0.0023, +0.0027]$	6/6	$[-0.0004, +0.0006]$	$[-0.0025, +0.0027]$	6/6
GASF	$[-0.0003, +0.0004]$	$[-0.0026, +0.0027]$	6/6	$[-0.0006, +0.0003]$	$[-0.0027, +0.0025]$	6/6
GADF	$[-0.0006, +0.0002]$	$[-0.0029, +0.0025]$	6/6	$[-0.0005, +0.0008]$	$[-0.0027, +0.0030]$	6/6
MTF	$[+0.0819, +0.1542]$	$[+0.0550, +0.1811]$	0/6	$[+0.0474, +0.1144]$	$[+0.0212, +0.1405]$	0/6

S10. Extended RP ablation

Table S10 isolates the RP decoder's geometric choices under the current metadata-free diagonal-centred inversion. All two-dimensional arms use the closed-form orientation, so the rows differ only in the stated geometric choice. On this sample, diagonal centring restores a predominantly two-dimensional Euclidean spectrum ($EVR_2 \approx 0.94$). Both the full state geometry and signed selection are necessary for near-lossless reconstruction, whereas inverting the PNG with the signal range instead of centring on the zero diagonal applies an affine shift that destroys the Euclidean structure.

Table S10: Controlled RP geometry ablation on 100 fold-0 windows. MAE is in m/s^2; EVR_2 is the fraction of positive MDS variance in the two leading dimensions.

Variant	MAE	Pearson	EVR_2
Metadata-free diagonal + signed selection	**0.009**	**1.000**	0.944
Incorrect signal-range inversion	1.259	0.941	0.218
Without signed selection	2.033	0.789	0.944
Legacy Jaccard one-dimensional baseline	2.160	0.919	0.729

The legacy graph-based row denotes an earlier internal one-dimensional heuristic and is included only as implementation provenance; it is not presented as a faithful reimplementation of any specific prior reconstruction paper.

S11. RP orientation: angular grid versus closed form

The orientation step of Section 4.2 was previously a search over a grid of $n_\theta = 720$ rotation angles crossed with the two reflections, keeping the eight lowest-overlap candidates for the signed comparison. It is now solved exactly: for each reflection parity the translation-corrected overlap residual is linear in $(\cos\theta, \sin\theta)$, so the minimiser is an eigenvector of a symmetric 2×2 matrix, giving four exact candidates. Table S11 compares the two on the same 150 activity-balanced windows (450 axis-series) used by the robustness block, with the metadata-free inversion in both arms so that only the orientation differs.

Three points follow. First, on the native float operator the closed form is exact to double precision (mean NMAE 9.1×10^{-17}; mean normalised squared-overlap diagnostic 3.2×10^{-31}), whereas the grid retains 3.5×10^{-4} — the discretisation error of a 0.5° spacing. This is the term the main text previously attributed to the RP algorithm. Second, the advantage survives 8-bit storage (0.00101 versus 0.00120) but is progressively masked by pixel noise, the two arms differing by under 3×10^{-4} from $\sigma = 2$ onwards. Third, total decoding takes 0.27 versus 0.60 s per axis-series in this finite-N Python implementation. Although MDS remains the asymptotically dominant $O(N^3)$ step, at $N = 127$ the interpreted loop over 1,440 grid/reflection candidates dominates wall-clock time; the closed form removes that loop.

The cycle residual is 0.0533 against 0.0512 on clean PNG and 0.1627 against 0.1596 at $\sigma = 8$ (closed form versus grid). These differences do not measure reconstruction quality: as shown in Section S5, the signed-RP cycle residual

is governed by sign flips at the $\cos(\pi/4)$ threshold, which a float-exact reconstruction also produces. The column is reported for completeness and no comparison is drawn from it.

The last column counts axis-series where the signed-RP criterion did not select the candidate with the lowest NMAE against the ground truth. The closed form never mis-selects on clean input, float or 8-bit. The grid's counts are large but not comparable in kind: its eight-candidate shortlist contains near-identical angles, so a different pick costs a mean NMAE regret of only 2.0×10^{-4} on clean 8-bit input rather than a wrong reflection or sign.

Table S11: RP orientation ablation on 150 activity-balanced fold-0 windows (450 axis-series; 900 for each noise level, two draws per window). Both arms use the metadata-free diagonal inversion. "Overlap res." is the normalised diagnostic returned after translation alignment; the closed form minimises its unnormalised quadratic numerator. "Sel. fails" counts axis-series where the signed criterion did not pick the NMAE-optimal candidate.

Condition	Orientation	NMAE mean	NMAE median	NMAE p95	NMAE max	Overlap res.	decode s/axis	Sel. fails
Native float	closed form	9.1×10^{-17}	6.9×10^{-17}	2.2×10^{-16}	6.7×10^{-16}	3.2×10^{-31}	0.283	0/450
	grid, $n_\theta = 720$	0.000352	0.000208	0.001230	0.003187	1.6×10^{-5}	0.601	98/450
Clean 8-bit PNG	closed form	0.001014	0.000826	0.001920	0.016522	8.3×10^{-5}	0.274	0/450
	grid, $n_\theta = 720$	0.001201	0.000941	0.002411	0.016506	1.1×10^{-4}	0.605	189/450
PNG $\sigma = 2$	closed form	0.004203	0.003300	0.008603	0.042689	9.0×10^{-4}	0.270	3/900
	grid, $n_\theta = 720$	0.004366	0.003484	0.008684	0.043733	9.8×10^{-4}	0.582	628/900
PNG $\sigma = 4$	closed form	0.008333	0.006407	0.017153	0.086463	3.5×10^{-3}	0.271	6/900
	grid, $n_\theta = 720$	0.008473	0.006576	0.016629	0.095938	3.6×10^{-3}	0.595	631/900
PNG $\sigma = 8$	closed form	0.017094	0.013117	0.040865	0.110892	1.42×10^{-2}	0.277	26/900
	grid, $n_\theta = 720$	0.017165	0.013152	0.041611	0.111118	1.43×10^{-2}	0.603	627/900
PNG $\sigma = 16$	closed form	0.034036	0.027319	0.073409	0.172416	5.41×10^{-2}	0.277	48/900
	grid, $n_\theta = 720$	0.034296	0.027582	0.074347	0.176806	5.31×10^{-2}	0.603	631/900

The historical full-corpus snapshot changes from MAE 0.0148 and MSE 0.0010 to $0.0142\,\mathrm{m/s^2}$ and $0.0008\,\mathrm{m^2/s^4}$, with Pearson unchanged at 0.99998. This is a regression check rather than a pure orientation ablation: the earlier snapshot also used operator-range metadata and a scale-dependent selector, whereas the refreshed run uses the production metadata-free path. The causal orientation comparison is therefore the controlled 150-window experiment above. Classification is insensitive at reporting precision: across the six classifiers and both protocols the largest reconstructed-accuracy change is below 0.001, and the regenerated per-class tables and equivalence verdicts of Table 7 are unaffected.

S12. Metadata-free RP decoding

The RP diagonal is identically zero, so under min–max storage the constant diagonal pixel level marks where $R = 0$; centring on it ($R = \mathrm{PNG} - \mathrm{median}_i\,\mathrm{PNG}_{ii}$) recovers the signed matrix up to a positive scale that classical MDS and the final range rescale absorb (Section 4.2). Table S12 compares, on 150 activity-balanced fold-0 windows and with the closed-form orientation in both arms, this metadata-free decoder with an operator-range oracle that inverts the PNG using the true per-window $(R_{\min}, R_{\max})$. The two methods receive the same two noise realisations per window, and those replicates are averaged within each window before summarisation. Their range-normalised errors and correlations are descriptively similar in both the clean and $\sigma = 8$ conditions. The metadata-free path has a slightly higher clean physical MAE (0.0135 vs $0.0109\,\mathrm{m/s^2}$); the gap is the price of not storing the operator range, and it is smaller than the pooled RP MAE of $0.0142\,\mathrm{m/s^2}$ reported for the adopted decoder over all 8,172 windows. No inferential equality claim is made from this ablation. The two arms share their noise draws with each other but not with the perturbation curve of Table 4, which is why the metadata-free entry here (0.0166) and the RP entry there (0.0167) differ in the fourth decimal.

Table S12: RP reconstruction with an operator-range oracle (true per-window $(R_{\min}, R_{\max})$) versus the adopted metadata-free diagonal-centred inverse, on 150 activity-balanced fold-0 windows, both using the closed-form orientation. At $\sigma = 8$, two paired noise draws per window are averaged within window.

Noise	Decoder	NMAE	MAE ($\mathrm{m/s^2}$)	Pearson
$\sigma = 0$	operator-range oracle	0.0008	0.0109	0.99999
	metadata-free (diagonal)	0.0010	0.0135	0.99998
$\sigma = 8$	operator-range oracle	0.0167	0.2272	0.99604
	metadata-free (diagonal)	0.0166	0.2256	0.99616

S13. GADF storage range and pivot sensitivity

Theoretical operator range and fixed PNG map. Under the restricted normalisation $\bar{x}_i \in [-\alpha, \alpha]$, the largest possible angular separation is $\pi - 2\arccos(\alpha)$. The corresponding GADF magnitude is therefore bounded by

$$\max_{i,j} |G^{\mathrm{D}}_{ij}| = \sin(\pi - 2\arccos\alpha) = 2\alpha\sqrt{1-\alpha^2}.$$

For $\alpha = 0.7$, this bound is approximately 0.9998, and every non-constant window attains it, because the restricted normalisation places one sample at $-\alpha$ and another at $+\alpha$. The encoder's per-window min–max map is therefore exactly the fixed map $[-0.9998, 0.9998] \to [0, 255]$, whereas the decoder inverts the image with $[-1, 1]$ and so applies a deterministic gain of 1.00013. The resulting discrepancy is about 2×10^{-4} at either endpoint, below one 8-bit intensity step. This is a deterministic consequence of the declared branch restriction, not an empirical estimate, and is why no per-window range metadata is needed.

Pivot-ensemble size. Averaging the reconstruction over the n_p most-central rows, rather than using a single pivot, is the second numerical choice. Table S13 reports a descriptive sweep on 1,500 windows drawn without replacement from the complete WISDM window set with seed 0. On this sample, pointwise error falls monotonically from a single pivot (0.0353) to five (0.0158, the fixed headline setting) and continues to 0.0119 at $n_p = 9$. The measured wall-clock differences are small because each additional pivot adds an $O(N)$ pass over an already-loaded matrix; these timings describe this implementation run and are not intended as a portable hardware benchmark. The sweep shows the effect of averaging but is not used to revise the already fixed headline configuration.

Table S13: Descriptive GADF pivot-ensemble sweep on 1 500 WISDM windows sampled without replacement with seed 0 (4 500 axis-series) at $\alpha = 0.7$. n_p is the number of averaged central pivot rows; MAE is the 8-bit PNG round-trip error in physical units; time is the observed wall-clock time per axis-series. The fixed headline setting is $n_p = 5$.

n_p	1	3	5	7	9
MAE (8-bit PNG)	0.0353	0.0206	**0.0158**	0.0134	0.0119
Time (ms/axis)	1.54	1.55	1.57	1.58	1.59

S14. Exploratory MTF sensitivity analyses

Controlled decoder comparison at fixed $Q = 4$. Table S14 compares the earlier cosine-pivot decoder, the Fiedler-ordering decoder, and two variants supplied with additional source-derived information. Every variant decodes the *identical* 8-bit image (the same uint8 PNG round trip), so the differences reflect the decoding algorithm alone and not storage precision. The stored values associated with the quantile bins are midpoints of the quantile boundaries, not within-bin sample medians. The optional sign bit records whether the first source sample lies above or below the source midrange and is used only to choose between the two global amplitude orientations. Neither form of side information is part of the headline MTF decoder contract.

Because all variants read the same stored image, the reduction from the cosine-pivot decoder (MAE 6.46) to Fiedler ordering (MAE 5.07) isolates the effect of spectral ordering: it is a genuine algorithmic improvement, not a storage-precision artefact. Adding source-derived within-bin information reduces the error further, but even with a per-bin midpoint and a sign bit the many-to-one quantisation loss keeps MTF (MAE 2.60) far above the continuous decoders (RP/GASF/GADF at ~ 0.01–0.02).

Table S14: Controlled MTF decoder comparison at $Q = 4$ on 100 randomly selected fold-0 windows (seed 0, $T = 128$). Every variant decodes the identical 8-bit image; MAE (m/s^2) and Pearson are averaged across axes.

Decoder variant	MAE	Pearson
Cosine-pivot heuristic	6.459	+0.036
Fiedler ordering	5.070	−0.002
Fiedler + boundary-midpoint metadata	3.942	+0.019
Fiedler + midpoint + sign bit	2.602	+0.465

The reductions after adding midpoint and orientation information illustrate which ambiguities these metadata can remove. They are not lower bounds on the error attainable by every possible MTF decoder, and the many-to-one loss within each quantile state remains. All headline MTF results use Fiedler ordering without midpoint or sign-bit metadata.

Bin-count sensitivity across Q. Table S15 sweeps Q over all 8,172 pooled windows. The pointwise MAE and Pearson are decoder-only, because every deterministic decoder reconstructs a given window identically regardless of its cross-validation assignment. The six right-hand columns add the regenerated downstream classifier sensitivity across Q: the mean accuracy on the MTF reconstructions at each Q, obtained by predicting with the fixed original-trained models under the window-level 3-fold protocol.

Table S15: MTF bin-count sensitivity over all pooled windows ($N = 8\,172$). Pointwise MAE (m/s^2) and Pearson are decoder-only (each window reconstructs identically regardless of fold); the KNN–CNN columns are mean downstream accuracy on the MTF reconstructions at each Q under the window-level 3-fold protocol, using the fixed original-trained models. Highlighting marks the headline $Q = 4$ and is descriptive, not nested model selection. Each row re-encodes and re-decodes every window at that Q, so the $Q = 4$ row is an independent recomputation and reproduces Table 2 to within $0.01\,m/s^2$ rather than exactly.

Q	MAE	Pearson	KNN	SVM	RF	LSTM	GRU	CNN
2	4.11	+0.002	0.490	0.705	0.762	0.756	0.771	0.774
4	**3.95**	**+0.013**	0.559	0.697	0.757	0.757	0.765	0.773
6	4.01	+0.004	0.556	0.697	0.757	0.755	0.761	0.770
8	4.03	−0.001	0.539	0.695	0.753	0.757	0.764	0.765
10	4.07	+0.000	0.527	0.693	0.754	0.756	0.762	0.758
16	4.13	−0.014	0.520	0.697	0.755	0.757	0.762	0.753

Both the pointwise error and the downstream accuracy vary little across Q: MAE stays within 3.95–4.13 and every classifier moves by at most a few points (KNN, the most Q-sensitive, drifts from 0.49 at $Q = 2$ to 0.52 at $Q = 16$; the neural models stay near 0.77). $Q = 4$ attains the lowest pointwise MAE and is at or near the best accuracy for most classifiers, so the headline choice is not a cherry-picked optimum. Because the sweep reuses the evaluation windows it is a sensitivity check rather than a nested selection, so we do not claim $Q = 4$ is optimal for unseen subjects; MTF remains a descriptive lossy control.

S15. RP–GADF structural diagnostic

S15.1. Null-Corrected RP/GADF Regional Overlap

The visual proximity between RP and GADF is easy to overstate from a single example, so we add a diagnostic test that deliberately separates spatial region alignment from marginal colour statistics. On the full set of non-overlapping smartwatch-accelerometer WISDM windows (window length $N = 128$), we encode the same signal as signed RP and GADF, align the matrices on the common 127×127 support, and quantise each image channel into B visual intensity ranges. For every matched pair $(\mathrm{RP}_i, \mathrm{GADF}_i)$, we compare the quantised regions against a conservative null pair $(\mathrm{RP}_i, \mathrm{GADF}_j)$, where $j \neq i$ is drawn from the same activity class. The primary statistic is a weighted regional Jaccard score, $\sum_b |R_b \cap G_b| / \sum_b |R_b \cup G_b|$, averaged across the grayscale view and the three RGB/axis channels. We also compute a distribution-only overlap that ignores pixel position and a boundary-F1 score between quantised region edges.

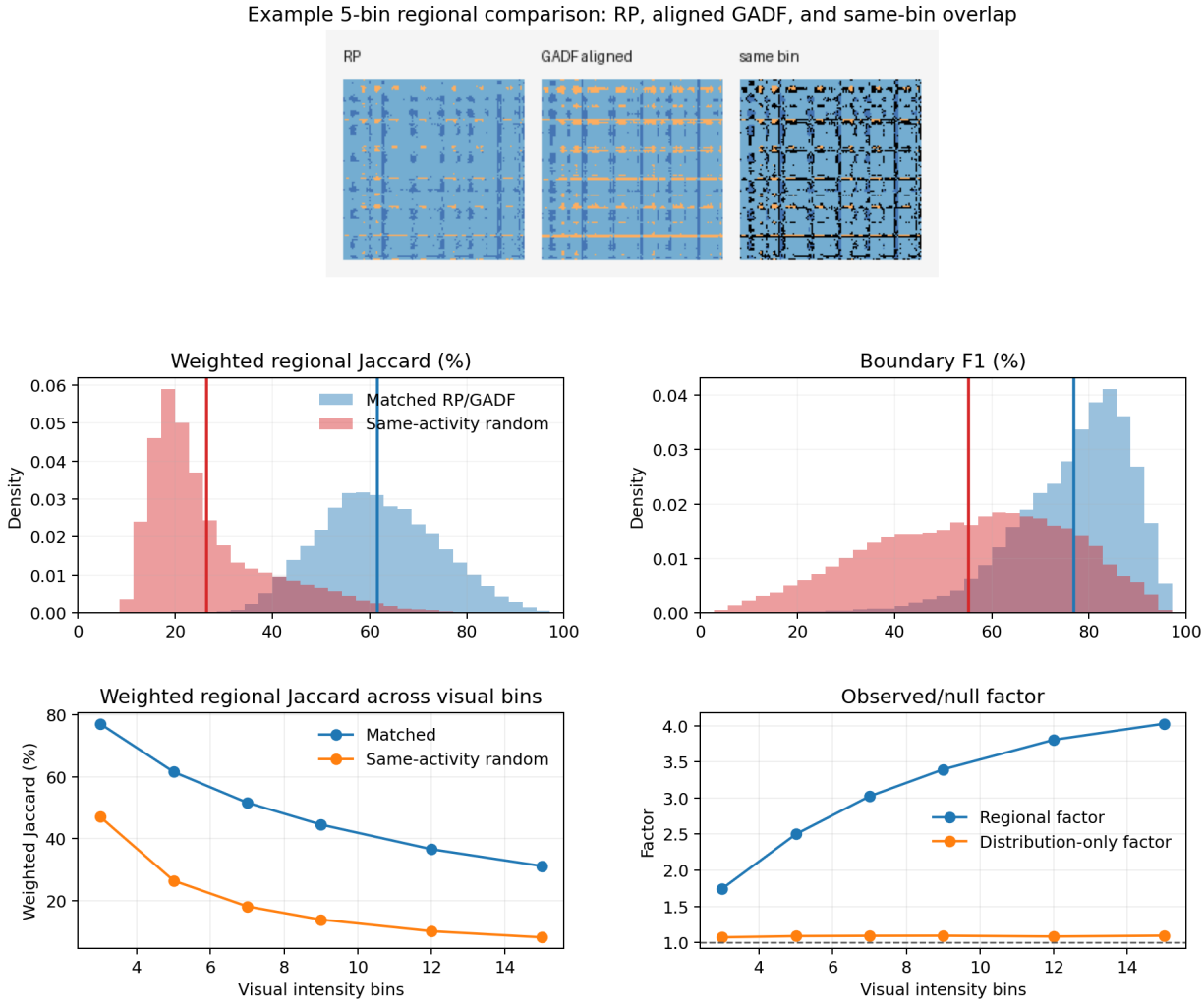


Figure S13: Null-corrected RP/GADF structural-overlap diagnostic. The top panel shows one 5-bin visual comparison: quantised RP, aligned quantised GADF, and pixels landing in the same bin. The middle histograms compare matched RP/GADF pairs against same-activity random pairings. The bottom row shows that matched regional overlap remains above the null baseline across independent visual quantisation levels $B \in \{3, 5, 7, 9, 12, 15\}$, while the distribution-only factor stays close to one.

Table S16: Null-corrected RP/GADF weighted regional overlap across visual intensity bins. Values are averaged over grayscale and the three axis channels. The null baseline pairs each RP image with a GADF image from a different window of the same activity class. Jaccard values are percentages; the Matched/null factor is the mean over the four channels of the per-channel matched/null ratio, so it is not the ratio of the two displayed column means.

Visual bins B	Matched Jaccard	Same-activity null	Matched–null gain	Matched/null factor
3	77.16	47.11	+30.05	1.75×
5	61.57	26.43	+35.14	2.50×
7	51.64	18.14	+33.50	3.02×
9	44.55	13.87	+30.68	3.40×
12	36.62	10.12	+26.50	3.81×
15	31.17	8.14	+23.03	4.03×

Table S16 and Figure S13 support a narrow descriptive claim: RP and GADF are not equivalent encodings, yet their coarse regions are more aligned when they come from the same signal window than under the same-activity pairing used here. As B increases, the absolute Jaccard decreases because the match becomes stricter, while the matched/null factor rises from 1.75× to 4.03×. This pattern is consistent with a spatial contribution beyond broad intensity histograms. The distribution-only factor stays near 1.07×–1.09× across the same sweep. Boundary F1 gives the same direction at coarse quantisation but becomes less diagnostic at high B because every image develops

many region edges. The diagnostic uses one declared null-pairing scheme and provides no subject-clustered inferential interval; it should therefore not be read as proof of operator equivalence or as a population-level hypothesis test.

S15.2. Structural relationship between signed RP and GADF

Operator-level comparison of pairwise structure. Before discussing the first-order amplitude expansion, it is useful to put signed RP and GADF in a common notation without claiming equality of their output spaces. Both can be written as pairwise functions of two-dimensional states,

$$M_{ij} \;=\; \omega(z_i, z_j),$$

where $z_i \in \mathbb{R}^2$ is method specific and $\omega : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$ is the associated pairwise operator. The two methods differ in both the state construction and the operator:

- RP under the Lu–Tong convention uses the sequence-dependent adjacent state $z_k = (x_k,\, x_{k+1})$ and a signed Euclidean distance, $\omega_{\mathrm{RP}}(u, v) = \sigma(u, v)\|u - v\|_2$, where the sign is assigned *externally* by the projection onto the reference direction $v_\star = (1, 1)^\top/\sqrt{2}$. With the thresholded sign rule used here, $\sigma(u, v)$ is not guaranteed to satisfy $\sigma(u, v) = -\sigma(v, u)$ for every pair, so the resulting matrix is directional but not necessarily antisymmetric.
- GADF uses the polar state $z_k = (\tilde{x}_k, \sqrt{1 - \tilde{x}_k^2}) = (\cos\phi_k, \sin\phi_k)$ and the operator $\omega_{\mathrm{GADF}}(u, v) = \det[v \mid u] = u_2v_1 - u_1v_2 = \sin(\phi_i - \phi_j)$, the signed area of the parallelogram spanned by the two states, whose sign is *intrinsic* to the operator and matches G^{D}_{ij} in Eq. (4).

The important consequence is therefore an analogy, not an identity. GADF lies *within* the $\binom{N}{2}$-dimensional space of antisymmetric zero-diagonal matrices by construction, because the determinant changes sign when its arguments are swapped; more specifically, its factorisation $G^{\mathrm{D}} = \mathbf{s}\,\mathbf{c}^\top - \mathbf{c}\,\mathbf{s}^\top$ (with $s_i = \sin\phi_i$, $c_i = \cos\phi_i$) shows it occupies only a rank-≤ 2 subvariety of that ambient space, not the whole of it. Signed RP has a zero diagonal and carries directional pairwise-distance information, but the Lu–Tong sign rule used in this paper does not force exact antisymmetry for nearly orthogonal displacements. The first-order expansion below quantifies why the two images can nevertheless look and behave similarly, while the structural differences between the two operators — GADF's intrinsic sign and its closed-form one-pass inverse, versus signed RP's externally assigned sign and its MDS-based reconstruction — follow from the determinant-based operator rather than from an exact shared matrix space.

Local first-order analogy. The visual proximity between RP and GADF can be examined through a local approximation without claiming a shared operator. GADF produces an antisymmetric zero-diagonal matrix, while signed RP produces a zero-diagonal directional distance matrix with related pairwise-difference content. For smooth signals whose local increments vary slowly (so that $x_{i+1} - x_i \approx x_{j+1} - x_j$ over the compared indices), both can be approximated by functions of the pairwise-difference matrix $\{x_i - x_j\}_{i,j}$. The signed RP under the Lu–Tong embedding satisfies

$$R_{ij} \;\approx\; \sigma(i, j)\,\sqrt{2}\,|x_i - x_j|,$$

where $\sigma(i, j)$ is the external orientation flag computed from the projection of $s_i - s_j$ onto $v = (1, 1)^\top/\sqrt{2}$. The GADF entry, expanded around a common normalized amplitude $\bar{x}$, satisfies

$$G^{\mathrm{D}}_{ij} \;=\; \sin(\phi_i - \phi_j) \;\approx\; \frac{\bar{x}_j - \bar{x}_i}{\sqrt{1 - \bar{x}^2}},$$

because $d(\arccos x)/dx = -1/\sqrt{1 - x^2}$, with both the samples $\bar{x}_i, \bar{x}_j$ and the expansion point $\bar{x}$ taken in the restricted $[-\alpha, \alpha]$ normalization of the GADF decoder. The sign of the GADF term is intrinsic to the trigonometric form rather than imposed externally. Thus, both expressions depend on pairwise amplitude differences at first order and both are zero on the diagonal, but only GADF is strictly antisymmetric under the stated construction. The restricted-branch GADF inverse is closed form because the angular interval $[\arccos(\alpha), \arccos(-\alpha)]$, with $\alpha = 0.7 < 1/\sqrt{2}$, keeps every angular difference on the principal branch and the image-derived anchor restores the global offset. Signed RP instead recovers a two-dimensional state geometry through MDS and uses overlap plus its external sign convention to choose an orientation. The approximation offers a plausible local explanation for visual similarity; it does not imply that the matrices contain identical information or admit the same inverse.

Practical interpretation. The operator analysis and the regional diagnostic are consistent with the empirical finding that both RP and GADF preserve enough directional pairwise structure for accurate controlled reconstruction, but they reach that result through different contracts. GASF supplies single-sample information on its diagonal; GADF supplies anchored angular differences under a restricted branch; and signed RP supplies adjacent-state distances plus an external orientation convention. The fidelity and equivalence results establish the usefulness of all three under those declared contracts. They do not establish a formal information equivalence among the encodings or a general preference outside the controlled reconstruction task.

A. Per-axis reconstruction errors

Table 2 in the main text reports the per-window global reconstruction error (averaged over the three accelerometer axes) for each encoder. This appendix gives the per-axis breakdown over the same $N = 8\,172$ pooled windows for RP (Table S17), MTF (Table S18), GASF (Table S19), and GADF (Table S20); in each, the Global column reproduces the per-window global figure of Table 2.

Table S17: RP per-axis reconstruction errors over all $N = 8\,172$ pooled windows (physical units, m/s^2); mean±std across windows. The Global column averages the three axes per window.

Metric	**x**	**y**	**z**	**Global**
Mean Absolute Error	0.0151 ± 0.0203	0.0153 ± 0.0201	0.0122 ± 0.0156	0.0142 ± 0.0156
Mean Squared Error	0.0010 ± 0.0075	0.0009 ± 0.0086	0.0006 ± 0.0027	0.0008 ± 0.0045
Pearson Correlation Coefficient	0.99998 ± 0.00009	0.99999 ± 0.00005	0.99998 ± 0.00006	0.99998 ± 0.00005

Table S18: MTF per-axis reconstruction errors over all $N = 8\,172$ pooled windows (physical units, m/s^2); mean±std across windows. The Global column averages the three axes per window.

Metric	**x**	**y**	**z**	**Global**
Mean Absolute Error	4.13 ± 5.11	4.18 ± 5.32	3.53 ± 4.66	3.94 ± 4.25
Mean Squared Error	57.39 ± 123.57	61.72 ± 157.60	47.18 ± 158.11	55.43 ± 113.46
Pearson Correlation Coefficient	-0.00214 ± 0.84525	0.03600 ± 0.83941	0.00680 ± 0.82844	0.01356 ± 0.48870

Table S19: GASF per-axis reconstruction errors over all $N = 8\,172$ pooled windows (physical units, m/s^2); mean±std across windows. The Global column averages the three axes per window.

Metric	**x**	**y**	**z**	**Global**
Mean Absolute Error	0.0236 ± 0.0243	0.0195 ± 0.0223	0.0183 ± 0.0216	0.0205 ± 0.0199
Mean Squared Error	0.0050 ± 0.0107	0.0048 ± 0.0127	0.0030 ± 0.0121	0.0043 ± 0.0088
Pearson Correlation Coefficient	0.99987 ± 0.00020	0.99990 ± 0.00024	0.99987 ± 0.00027	0.99988 ± 0.00014

Table S20: GADF per-axis reconstruction errors over all $N = 8\,172$ pooled windows (physical units, m/s^2); mean±std across windows. The Global column averages the three axes per window.

Metric	**x**	**y**	**z**	**Global**
Mean Absolute Error	0.0159 ± 0.0189	0.0169 ± 0.0215	0.0144 ± 0.0199	0.0157 ± 0.0167
Mean Squared Error	0.0008 ± 0.0020	0.0009 ± 0.0028	0.0007 ± 0.0028	0.0008 ± 0.0018
Pearson Correlation Coefficient	0.99999 ± 0.00002	0.99999 ± 0.00002	0.99999 ± 0.00002	0.99999 ± 0.00001

B. Per-Class Classification Results

This appendix reports the complete per-class precision, recall, and F1-score results for the six evaluated classifiers. The main text retains only the compact accuracy comparison required to interpret the differences among reconstruction methods. Per-class Support values are per-class fold-means rounded to integers and may sum to 2723 rather than the 2724 windows per fold reported in the Accuracy row (independent rounding).

Table S21: Classification Results for GRU on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.89±0.03	0.93±0.01	0.91±0.01	544	0.89±0.03	0.93±0.01	0.91±0.01	544	0.65±0.01	0.61±0.04	0.63±0.01	544	0.89±0.03	0.93±0.01	0.91±0.01	544	0.89±0.03	0.93±0.01	0.91±0.01	544
1	0.99±0.01	0.99±0.00	0.99±0.00	532	0.99±0.01	0.99±0.00	0.99±0.00	532	0.75±0.02	0.99±0.00	0.85±0.01	532	0.99±0.01	0.99±0.00	0.99±0.00	532	0.99±0.01	0.99±0.00	0.99±0.00	532
2	0.91±0.01	0.86±0.03	0.88±0.01	536	0.91±0.01	0.86±0.03	0.88±0.01	536	0.57±0.01	0.52±0.03	0.54±0.01	536	0.91±0.01	0.86±0.03	0.88±0.01	536	0.91±0.01	0.86±0.03	0.88±0.01	536
3	0.88±0.01	0.96±0.01	0.92±0.00	551	0.88±0.01	0.96±0.01	0.92±0.00	551	0.90±0.01	0.90±0.01	0.90±0.01	551	0.88±0.01	0.96±0.01	0.92±0.00	551	0.88±0.01	0.96±0.01	0.92±0.00	551
4	0.95±0.01	0.89±0.01	0.92±0.00	560	0.95±0.01	0.89±0.01	0.92±0.00	560	0.96±0.01	0.81±0.01	0.88±0.00	560	0.95±0.01	0.89±0.01	0.92±0.00	560	0.95±0.01	0.89±0.01	0.92±0.00	560
Accuracy	0.92±0.00 (2724)				0.92±0.00 (2724)				0.77±0.01 (2724)				0.92±0.00 (2724)				0.92±0.00 (2724)			
Macro Avg	0.93±0.00/0.92±0.00/0.92±0.00				0.93±0.00/0.92±0.00/0.92±0.00				0.77±0.01/0.76±0.01/0.76±0.01				0.92±0.00/0.92±0.00/0.92±0.00				0.93±0.00/0.92±0.00/0.92±0.00			
Weighted Avg	0.93±0.00/0.92±0.00/0.92±0.00				0.93±0.00/0.92±0.00/0.92±0.00				0.77±0.01/0.77±0.01/0.76±0.01				0.92±0.00/0.92±0.00/0.92±0.00				0.93±0.00/0.92±0.00/0.92±0.00			

Table S22: Classification Results for KNN on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.74±0.01	0.52±0.01	0.61±0.01	544	0.74±0.01	0.52±0.01	0.61±0.01	544	0.64±0.01	0.49±0.02	0.55±0.01	544	0.74±0.01	0.52±0.01	0.61±0.01	544	0.74±0.01	0.52±0.01	0.61±0.01	544
1	1.00±0.00	0.75±0.01	0.86±0.01	532	1.00±0.00	0.75±0.01	0.86±0.01	532	0.99±0.02	0.37±0.00	0.54±0.00	532	1.00±0.00	0.75±0.01	0.86±0.01	532	1.00±0.00	0.75±0.01	0.86±0.01	532
2	0.49±0.04	0.04±0.01	0.08±0.02	536	0.49±0.04	0.04±0.01	0.08±0.02	536	0.43±0.04	0.07±0.00	0.12±0.00	536	0.50±0.04	0.04±0.01	0.08±0.02	536	0.50±0.04	0.04±0.01	0.08±0.01	536
3	0.79±0.00	0.96±0.00	0.87±0.00	551	0.79±0.00	0.96±0.00	0.87±0.00	551	0.63±0.01	0.94±0.01	0.75±0.01	551	0.79±0.00	0.96±0.00	0.87±0.00	551	0.79±0.00	0.96±0.00	0.87±0.00	551
4	0.42±0.00	0.92±0.01	0.58±0.00	560	0.42±0.00	0.92±0.01	0.58±0.00	560	0.42±0.01	0.90±0.01	0.57±0.01	560	0.42±0.00	0.92±0.01	0.58±0.00	560	0.42±0.00	0.92±0.01	0.58±0.00	560
Accuracy	0.64±0.00 (2724)				0.64±0.00 (2724)				0.56±0.01 (2724)				0.64±0.00 (2724)				0.64±0.00 (2724)			
Macro Avg	0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00				0.62±0.01/0.55±0.01/0.51±0.01				0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00			
Weighted Avg	0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00				0.62±0.01/0.56±0.01/0.51±0.01				0.69±0.01/0.64±0.00/0.60±0.00				0.69±0.01/0.64±0.00/0.60±0.00			

Table S23: Classification Results for CNN on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.89±0.05	0.89±0.05	0.89±0.01	544	0.89±0.05	0.89±0.05	0.89±0.01	544	0.61±0.07	0.73±0.08	0.66±0.01	544	0.89±0.05	0.89±0.05	0.89±0.01	544	0.89±0.05	0.89±0.05	0.89±0.01	544
1	0.99±0.01	0.98±0.00	0.99±0.00	532	0.99±0.01	0.98±0.00	0.99±0.00	532	0.88±0.03	0.97±0.02	0.92±0.01	532	0.99±0.01	0.98±0.00	0.99±0.00	532	0.99±0.01	0.98±0.00	0.99±0.00	532
2	0.88±0.04	0.87±0.05	0.87±0.01	536	0.88±0.04	0.87±0.05	0.87±0.01	536	0.62±0.03	0.55±0.10	0.57±0.05	536	0.88±0.04	0.87±0.05	0.87±0.01	536	0.88±0.04	0.87±0.05	0.87±0.01	536
3	0.84±0.05	0.97±0.02	0.90±0.02	551	0.84±0.05	0.97±0.02	0.90±0.02	551	0.85±0.04	0.89±0.04	0.87±0.01	551	0.84±0.04	0.97±0.02	0.90±0.02	551	0.84±0.05	0.97±0.02	0.90±0.02	551
4	0.97±0.02	0.83±0.05	0.89±0.02	560	0.97±0.02	0.82±0.05	0.89±0.02	560	0.97±0.03	0.73±0.04	0.83±0.02	560	0.97±0.02	0.83±0.05	0.89±0.02	560	0.97±0.02	0.82±0.05	0.89±0.02	560
Accuracy	0.91±0.00 (2724)				0.91±0.00 (2724)				0.77±0.01 (2724)				0.91±0.01 (2724)				0.91±0.00 (2724)			
Macro Avg	0.91±0.00/0.91±0.00/0.91±0.00				0.91±0.00/0.91±0.00/0.91±0.00				0.79±0.01/0.77±0.01/0.77±0.01				0.91±0.00/0.91±0.01/0.91±0.01				0.91±0.00/0.91±0.00/0.91±0.00			
Weighted Avg	0.91±0.00/0.91±0.00/0.91±0.01				0.91±0.00/0.91±0.00/0.91±0.00				0.79±0.01/0.77±0.01/0.77±0.01				0.91±0.00/0.91±0.01/0.91±0.01				0.91±0.00/0.91±0.00/0.91±0.00			

Table S24: Classification Results for Random Forest on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.85±0.01	0.79±0.00	0.82±0.00	544	0.85±0.01	0.79±0.00	0.82±0.01	544	0.68±0.01	0.60±0.01	0.64±0.00	544	0.85±0.01	0.79±0.01	0.82±0.01	544	0.85±0.01	0.79±0.00	0.82±0.00	544
1	0.94±0.00	0.99±0.00	0.96±0.00	532	0.94±0.00	0.99±0.00	0.96±0.00	532	0.65±0.01	1.00±0.00	0.79±0.01	532	0.94±0.00	0.99±0.00	0.96±0.00	532	0.94±0.00	0.99±0.00	0.96±0.00	532
2	0.79±0.00	0.82±0.01	0.81±0.01	536	0.79±0.00	0.82±0.01	0.80±0.01	536	0.63±0.01	0.51±0.02	0.56±0.01	536	0.79±0.01	0.82±0.01	0.80±0.01	536	0.79±0.00	0.82±0.01	0.80±0.01	536
3	0.92±0.01	0.95±0.00	0.93±0.01	551	0.92±0.01	0.95±0.00	0.93±0.01	551	0.93±0.01	0.89±0.01	0.91±0.00	551	0.92±0.01	0.95±0.00	0.93±0.01	551	0.92±0.01	0.95±0.00	0.93±0.00	551
4	0.96±0.00	0.90±0.01	0.93±0.00	560	0.96±0.00	0.90±0.01	0.93±0.00	560	0.94±0.01	0.79±0.01	0.86±0.01	560	0.96±0.00	0.90±0.01	0.93±0.00	560	0.96±0.00	0.90±0.01	0.93±0.00	560
Accuracy	0.89±0.00 (2724)				0.89±0.00 (2724)				0.76±0.00 (2724)				0.89±0.00 (2724)				0.89±0.00 (2724)			
Macro Avg	0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.77±0.00/0.76±0.00/0.75±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00			
Weighted Avg	0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.77±0.00/0.76±0.00/0.75±0.00				0.89±0.00/0.89±0.00/0.89±0.00				0.89±0.00/0.89±0.00/0.89±0.00			

Table S25: Classification Results for Support Vector Machine on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.76±0.01	0.65±0.02	0.70±0.01	544	0.76±0.01	0.65±0.01	0.70±0.01	544	0.52±0.01	0.71±0.02	0.60±0.01	544	0.76±0.01	0.65±0.01	0.70±0.01	544	0.76±0.00	0.65±0.01	0.70±0.01	544
1	0.97±0.01	0.97±0.00	0.97±0.00	532	0.97±0.01	0.97±0.00	0.97±0.00	532	0.69±0.01	1.00±0.00	0.81±0.01	532	0.97±0.01	0.97±0.00	0.97±0.00	532	0.97±0.01	0.97±0.00	0.97±0.00	532
2	0.66±0.01	0.69±0.01	0.67±0.01	536	0.65±0.01	0.69±0.01	0.67±0.01	536	0.49±0.04	0.17±0.01	0.25±0.02	536	0.65±0.01	0.69±0.01	0.67±0.01	536	0.66±0.01	0.69±0.01	0.67±0.01	536
3	0.84±0.01	0.84±0.00	0.84±0.00	551	0.85±0.01	0.84±0.00	0.84±0.00	551	0.86±0.01	0.81±0.01	0.83±0.00	551	0.84±0.01	0.84±0.00	0.84±0.00	551	0.84±0.01	0.84±0.00	0.84±0.00	551
4	0.78±0.01	0.85±0.01	0.81±0.00	560	0.78±0.01	0.85±0.01	0.81±0.01	560	0.86±0.00	0.81±0.00	0.83±0.00	560	0.78±0.01	0.85±0.01	0.81±0.00	560	0.78±0.00	0.85±0.01	0.81±0.00	560
Accuracy	0.80±0.00 (2724)				0.80±0.00 (2724)				0.70±0.01 (2724)				0.80±0.00 (2724)				0.80±0.00 (2724)			
Macro Avg	0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.00/0.80±0.00/0.80±0.00				0.69±0.01/0.70±0.01/0.67±0.01				0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.00/0.80±0.00/0.80±0.00			
Weighted Avg	0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.00/0.80±0.00/0.80±0.00				0.69±0.01/0.70±0.01/0.67±0.01				0.80±0.01/0.80±0.00/0.80±0.01				0.80±0.00/0.80±0.00/0.80±0.00			

Table S26: Classification Results for LSTM on the stratified window-level 3-fold protocol (mean ± std across folds 0, 1, 2). Per-class precision/recall/F1 and support averaged over the same three folds; footer accuracies averaged likewise.

	Original				RP				MTF				GASF				GADF			
Class	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp	Prec	Rec	F1	Supp
0	0.80±0.11	0.85±0.06	0.81±0.04	544	0.80±0.11	0.85±0.06	0.81±0.04	544	0.62±0.06	0.65±0.10	0.63±0.02	544	0.79±0.10	0.85±0.06	0.81±0.04	544	0.80±0.11	0.85±0.07	0.81±0.04	544
1	0.98±0.02	0.98±0.01	0.98±0.01	532	0.98±0.02	0.98±0.01	0.98±0.01	532	0.77±0.01	0.96±0.02	0.85±0.02	532	0.98±0.02	0.98±0.01	0.98±0.01	532	0.98±0.02	0.98±0.01	0.98±0.01	532
2	0.83±0.05	0.71±0.21	0.74±0.12	536	0.83±0.05	0.71±0.21	0.74±0.12	536	0.61±0.03	0.46±0.18	0.50±0.13	536	0.83±0.04	0.71±0.20	0.74±0.12	536	0.83±0.05	0.71±0.21	0.74±0.12	536
3	0.87±0.01	0.94±0.01	0.90±0.00	551	0.87±0.01	0.94±0.01	0.90±0.00	551	0.87±0.00	0.92±0.00	0.89±0.00	551	0.87±0.00	0.94±0.01	0.90±0.00	551	0.87±0.01	0.94±0.01	0.90±0.00	551
4	0.92±0.04	0.86±0.02	0.89±0.01	560	0.92±0.03	0.86±0.02	0.89±0.01	560	0.95±0.01	0.79±0.02	0.86±0.01	560	0.92±0.03	0.86±0.02	0.89±0.01	560	0.92±0.03	0.86±0.02	0.89±0.01	560
Accuracy	0.87±0.03 (2724)				0.87±0.03 (2724)				0.76±0.02 (2724)				0.87±0.03 (2724)				0.87±0.03 (2724)			
Macro Avg	0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04				0.76±0.01/0.76±0.02/0.75±0.03				0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04			
Weighted Avg	0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04				0.76±0.01/0.76±0.02/0.75±0.03				0.88±0.02/0.87±0.03/0.86±0.04				0.88±0.02/0.87±0.03/0.86±0.04			